

CENTRAL CALIFORNIA
EMERGENCY MEDICAL SERVICES
A Division of the Fresno County Department of Public Health

Manual	Emergency Medical Services Administrative Policies and Procedures	Policy Number 530.31
Subject	Paramedic Treatment Protocols	Page 1 of 3
NEONATAL RESUSCITATION (PATIENTS LESS THAN 24 HOURS OF AGE)		
References	Title 22, Division 9, Chapter 3.3 of the California Code of Regulations	Effective Fresno County: 01/15/82 Kings County: 04/10/89 Madera County: 06/15/85 Tulare County: 04/19/05

STANDING ORDERS	
Warm and prevent hypothermia	Dry the infant, turn up heat in ambulance, apply cap.
Airway	Suction with bulb syringe only if secretions are obstructing the airway. Routine suctioning in the absence of airway obstruction is not indicated. Suction mouth first, then nostrils.
Stimulate	Provide tactile stimulation by drying the infant and rubbing the back and soles of the feet if respiratory effort is inadequate.
Assess heart rate	Initial HR assessment can be made by palpating the pulse in the umbilical stump or auscultating the chest.
Monitor	Place patient on cardiac monitor for continuous heart rate monitoring if ongoing resuscitation is required. Place O2 sat probe on the right hand.
If heart rate < 100 beats/min, gasping, or apnea	
Ventilate	Ventilate with 40-60 ventilations per minute on room air for 30 seconds. Use only the minimum volume necessary to achieve chest rise.
Reassess heart rate after 30 seconds	<u>HR ≥ 100</u> : Prevent hypothermia and continuously reassess need for resuscitation. <u>HR 60-100</u> : Reposition airway, evaluate airway patency and need for suction, evaluate for chest rise. Continue ventilations at 40-60 ventilations/min. <u>HR < 60</u> : Initiate chest compressions
Oxygen	Convert to ventilations with high flow O2 for persistent bradycardia, apnea, or central cyanosis despite initial ventilation on room air.

STANDING ORDERS – CONTINUED ON NEXT PAGE

Approved By	Daniel J. Lynch	Revision
EMS Director	(Signature on File at EMS Agency)	09/01/2025
EMS Medical Director	Miranda Lewis, MD (Signature on File at EMS Agency)	

Subject	Paramedic Treatment Protocols – Neonatal Resuscitation	Policy Number 530.31
---------	--	----------------------

STANDING ORDERS (CONTINUED)	
If heart rate < 60 beats/minute	
Chest Compressions	<u>If HR remains < 60 despite 30 seconds of ventilation</u> : Initiate chest compressions at compression to ventilation ratio of 3:1 (total of 120 events per minute).
Reassess heart rate after 60 seconds	<u>HR > 100</u> : Prevent hypothermia and continuously reassess need for resuscitation. <u>HR 60-100</u> : Continue with positive pressure ventilation <u>HR < 60</u> : Continue CPR. Initiate IO access and epinephrine
Vascular access	IO is preferred.
Epinephrine	<u>If HR persistently < 60 despite one minute of chest compressions and ventilations</u> : - Epinephrine 1:10,000 0.01 mg/kg (0.1 ml/kg) IV/IO. Repeat every 3-5 minutes while HR < 60 persists up to 5 doses. Contact base for additional doses.
Normal saline	<u>If HR persistently < 60 despite CPR and epinephrine and blood loss is suspected</u> : - Administer 10 ml/kg normal saline IV/IO. May repeat x 1.
Transport	
STAT Transport	Begin early transport after initial steps of resuscitation and assessment are complete. Notify hospital of ETA when enroute.
Contact Base	Per EMS Policy #530.02

SPECIAL CONSIDERATION AND PRIORITIES

- Most newly born infants do not require resuscitation. If the infant is full term, has good tone, and is crying they should be placed on the mother's chest skin-to-skin while ongoing evaluation and monitoring takes place. Attention should be given to providing a warm environment and coverings to prevent neonatal hypothermia.
- A rise in heart rate is the most important indicator of effective ventilation and response to resuscitation. Continuous heart rate monitoring is critical in guiding the next steps of resuscitation.
- Perform chest compressions with both thumbs (with hands encircling the back), at the mid-sternum and intermammary line, at 1/3" – 1/2" depth of chest.
- An appropriate infant sized pulse oximeter probe placed on the right palm/wrist should be used to monitor SpO₂. Hypoxia is expected in the first minutes after birth (see table). Excessive oxygen contributes to worse outcomes. High flow O₂ should be reserved for patients with persistent bradycardia and those requiring CPR.

Target SpO ₂ after birth	
1 min	60-65%
2 min	65-70%
3 min	70-75%
4 min	75-80%
5 min	80-85%
10 min	85-90%
- Transport an arrested neonate lights/siren to the nearest appropriate hospital. Do not delay transport if difficulty with IV/IO access. Priorities should be good CPR focusing on quality ventilations and rapid transport once the initial steps of resuscitation have been completed on scene.
- Consider early Base Hospital contact for critically ill patients if manpower allows. Do not delay initial treatment to make Base Hospital contact.
- Neonates with confirmed gestational age less than 20 weeks are considered non-viable and resuscitation is not indicated. Assessment of gestational age based on physical characteristics alone is challenging and unreliable. If there are concerns for viability of the infant based on size or exam findings and gestational age is unclear, initiate resuscitation and contact base early for guidance.

Subject Paramedic Treatment Protocols – Neonatal Resuscitation	Policy Number 530.31
---	-------------------------

8. Use Broselow tape to estimate weight. The weight of an average full-term newborn is around 3 kg.

Epinephrine Dosing Chart Concentration 1:10,000 (0.1mg/1ml)		
Weight	Dose in mg	Volume in ml
2 kg	0.02 mg	0.2 ml
3 kg	0.03 mg	0.3 ml
4 kg	0.04 mg	0.4 ml
5 kg	0.05 mg	0.5 ml